

Movie Trivia QA System - Summary

System Overview

The *Movie Trivia QA* system is a retrieval-augmented question answering (QA) tool developed to answer user queries about movies. It integrates a local movie trivia dataset with a vector search mechanism and utilizes a language model (gpt-4o) for generating context-aware answers. In cases where the system cannot find relevant information locally, it falls back to a web search mechanism to enhance coverage.

Workflow

1. **User inputs a question** via the interface (Streamlit).
 2. The question is **embedded** into a vector using Sentence Transformers.
 3. FAISS **searches** for the most relevant movie trivia chunks.
 4. Retrieved chunks are **combined as context** and sent to gpt-4o.
 5. gpt-4o **generates an answer** based on the context.
 6. If gpt-4o replies with "I don't know", the system:
 - Performs a **web search** using SerpAPI
 - Sends web search results as context to gpt-4o
 7. The final answer is **displayed to the user**.
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Key Components

- **Movie Trivia Dataset:** JSON file with structured movie information
- **Embedding Model:** `sentence-transformers/all-MiniLM-L6-v2`
- **Vector Store:** FAISS index built from movie trivia embeddings

- **Language Model:** OpenAI GPT-4o for answer generation
 - **Web Search:** SerpAPI Google Search for fallback queries
 - **Interface:** Streamlit app or Jupyter notebook for interaction
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Current Capabilities

- Answers questions about movie titles, plots, actors, directors, and years.
 - Uses semantic search to find relevant information.
 - Falls back to web search when local data is insufficient.
 - Provides clear "I don't know" responses when no answer is found.
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Limitations & Future Work

- No query intent detection - system attempts to answer all questions regardless of relevance.
 - No confidence-based fallback - fallback triggered only by explicit "I don't know" in GPT output.
 - Web search results are used without advanced filtering or summarization.
 - Each OpenAI or web search call is made fresh; repeated queries are not cached, which may slow down responses.
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